INF10025 Tutorial 06 Answers

Complete Solutions & Explanations for Week 6

Question 1: Foreign Keys in Relational Schema

a. TASK relation: 1 Foreign Key (`ProjectId`).
b. PROJECT relation: 1 Foreign Key (`LeaderId`).

Question 2: PROJECT Relation Constraints

a. Foreign Key Name: `LeaderId`
b. Allowed NULL? Yes, unless specified as `NOT NULL` by mandatory business rules.

Question 3: TASK Relation Constraints

a. Foreign Key Name: `ProjectId`
b. Allowed NULL? Yes, unless specified as `NOT NULL` by mandatory business rules.

Question 4: Table Creation Sequence

Parent tables must be created prior to child tables referencing them:
a. First: `LEADER` (Parent table with no FK)
b. Second: `PROJECT` (Child of LEADER, Parent of TASK)
c. Last: `TASK` (Child of PROJECT)

Question 5 & 6: Parent and Child Rows Defined

Parent Row: A row in a primary table containing a Primary Key value referenced by Foreign Keys in other tables.
Child Row: A row in a related table containing a Foreign Key value referencing a Parent row's Primary Key.

Question 7: Parent-Child True/False

a. Leader is a parent of Project: **True**
b. Task is a parent of Project: **False** (Project is parent of Task)
c. Leader is a parent of Task: **False** (Indirect relationship)
d. Task is a parent of Leader: **False**

Question 8: Oracle Deletion Rules (Referential Integrity)

a. Parent row with no child rows deleted without error: **True**
b. Child row with parent row deleted without error: **True**
c. Child row may not be orphaned: **True**
d. Parent row cannot exist without having child rows: **False**
e. Parent row that has child rows deleted without error: **False** (Triggers `ORA-02292: child record found` error)

Question 9: Person Table WHERE Query Results

a. WHERE QtyA+QtyB < 10 → **5 rows** (Susan, Dick, Selina, Sonita, Dave)
b. WHERE QtyA >= 5 AND QtyB < 5 → **2 rows** (Eric Twinge, Dick Grayson)
c. WHERE NOT (QtyA >= 5 AND QtyB < 5) → **8 rows** (All except Eric & Dick; excludes NULLs)
d. WHERE gender = 'F' OR gender = 'M' AND QtyA+QtyB < 5 → **5 rows** (All 5 females)
e. WHERE NOT (QtyA + QtyB) < 10 AND Gender = 'F' → **2 rows** (Diana Prince, Barbara Gordon)
f. WHERE UPPER(Name) Like '%RI%' OR UPPER(Name) Like '%IN%' → **4 rows** (Eric, Diana, Dick, Rick)
g. WHERE LOWER(Name) Like '%or%' → **3 rows** (Johnny Storm, Barbara Gordon, Dave Orkle)
h. WHERE UPPER(Name) Like '%SON%' → **2 rows** (Dick Grayson, Sonita Marks)
i. WHERE Gender = 'M' OR QtyA >= 5 AND Gender = 'F' OR QtyB < 2 → **8 rows**
j. WHERE QtyA > 2 AND Gender = 'F' OR Gender = 'M' → **10 rows**
k. WHERE IdNo > 0 AND IdNo < 20 OR QtyB < 5 → **7 rows**
l. WHERE Gender = 'M' OR QtyA > 4 AND Gender = 'F' OR QtyB < 5 → **9 rows**
m. WHERE Gender = 'M' AND Gender = 'F' OR QtyB > 0 AND QtyB < 5 → **4 rows**

Question 10: SQL INNER JOIN Query (Project & Leader)

```
SELECT P.Title, L.Name
FROM PROJECT P
INNER JOIN LEADER L ON P.LeaderId = L.LeaderId;
```

Question 11: SQL Multi-table JOIN Query

```
SELECT T.Description, T.Cost, P.Title, L.Name
FROM TASK T
INNER JOIN PROJECT P ON T.ProjectId = P.ProjectId
INNER JOIN LEADER L ON P.LeaderId = L.LeaderId;
```